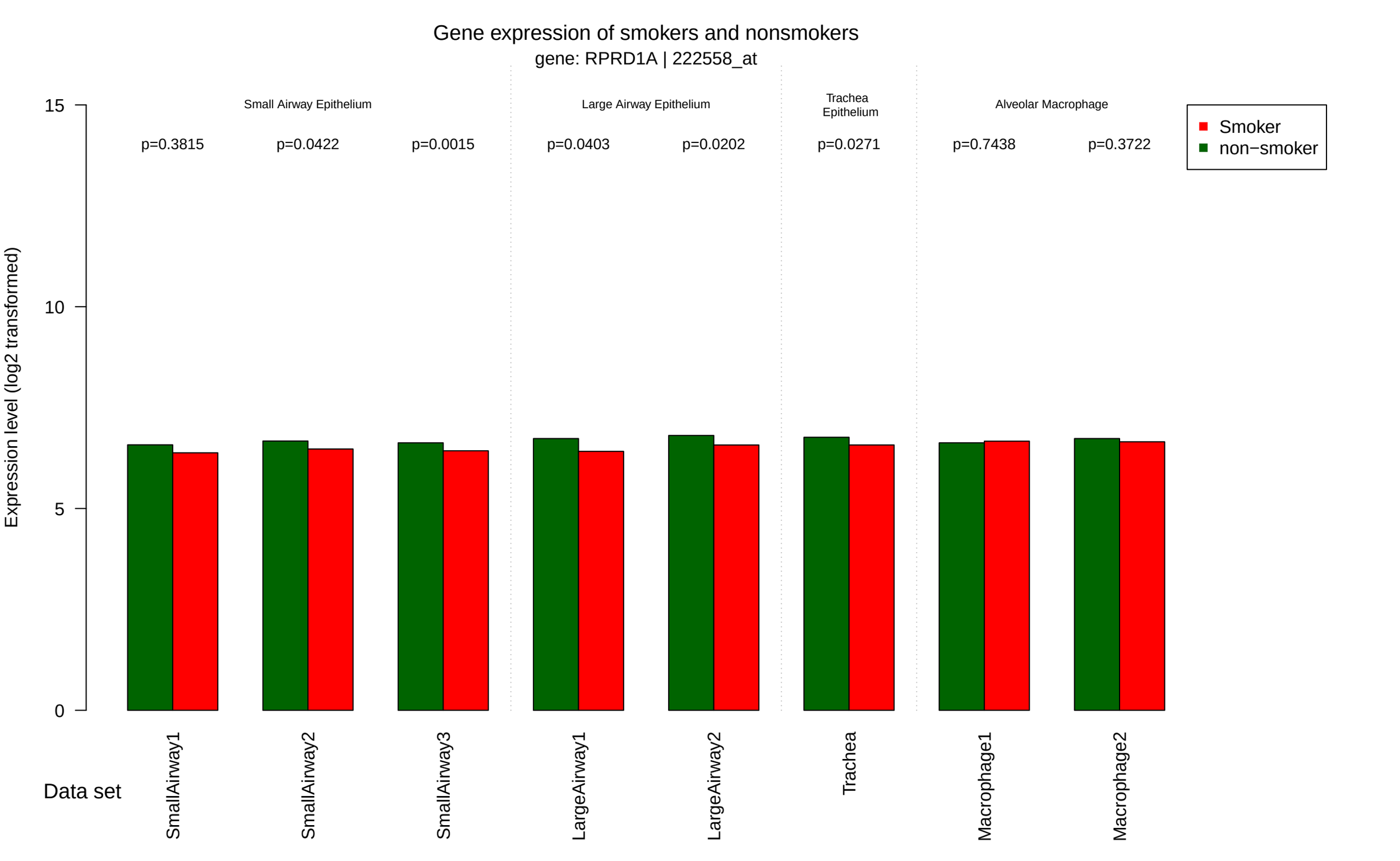
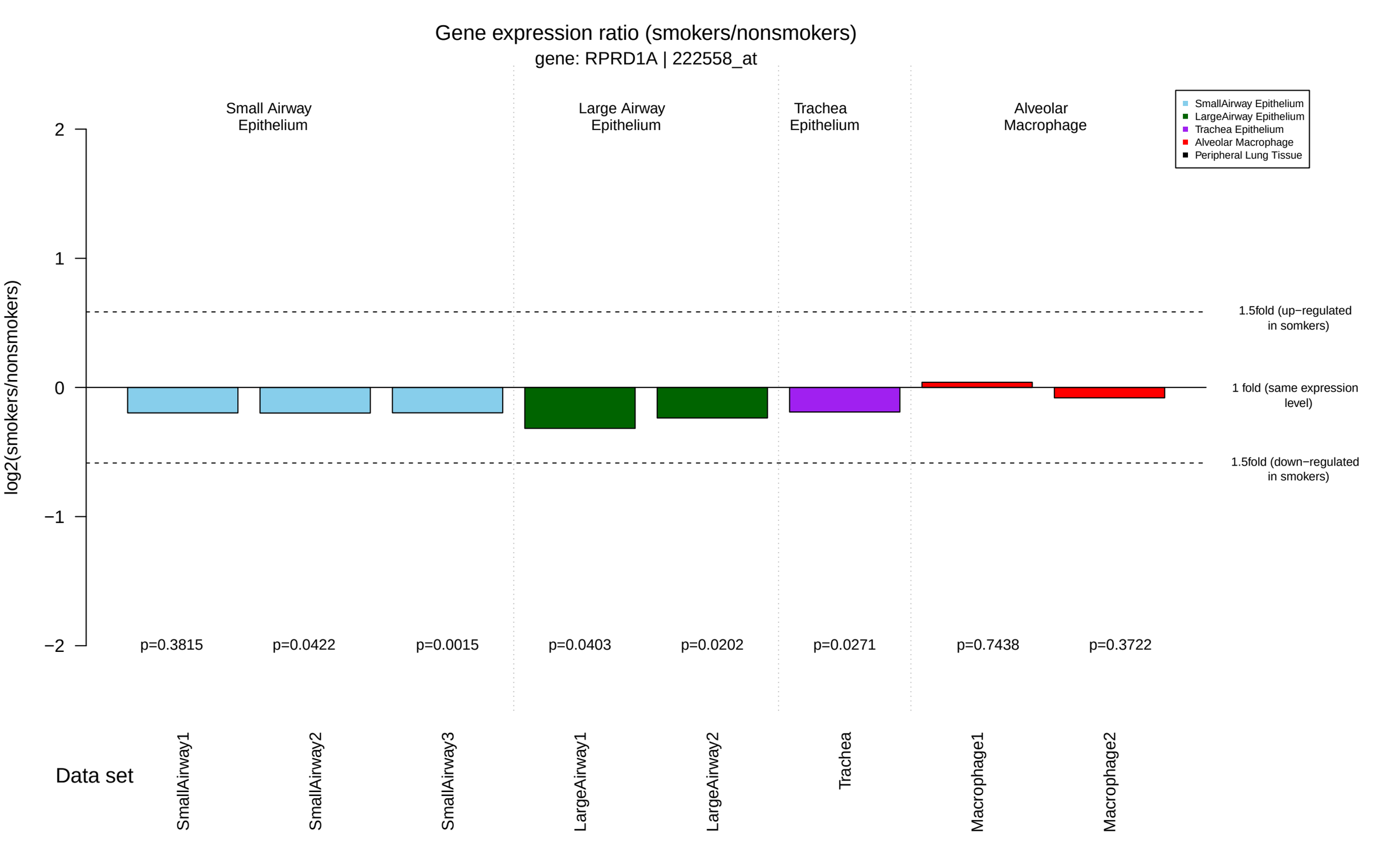
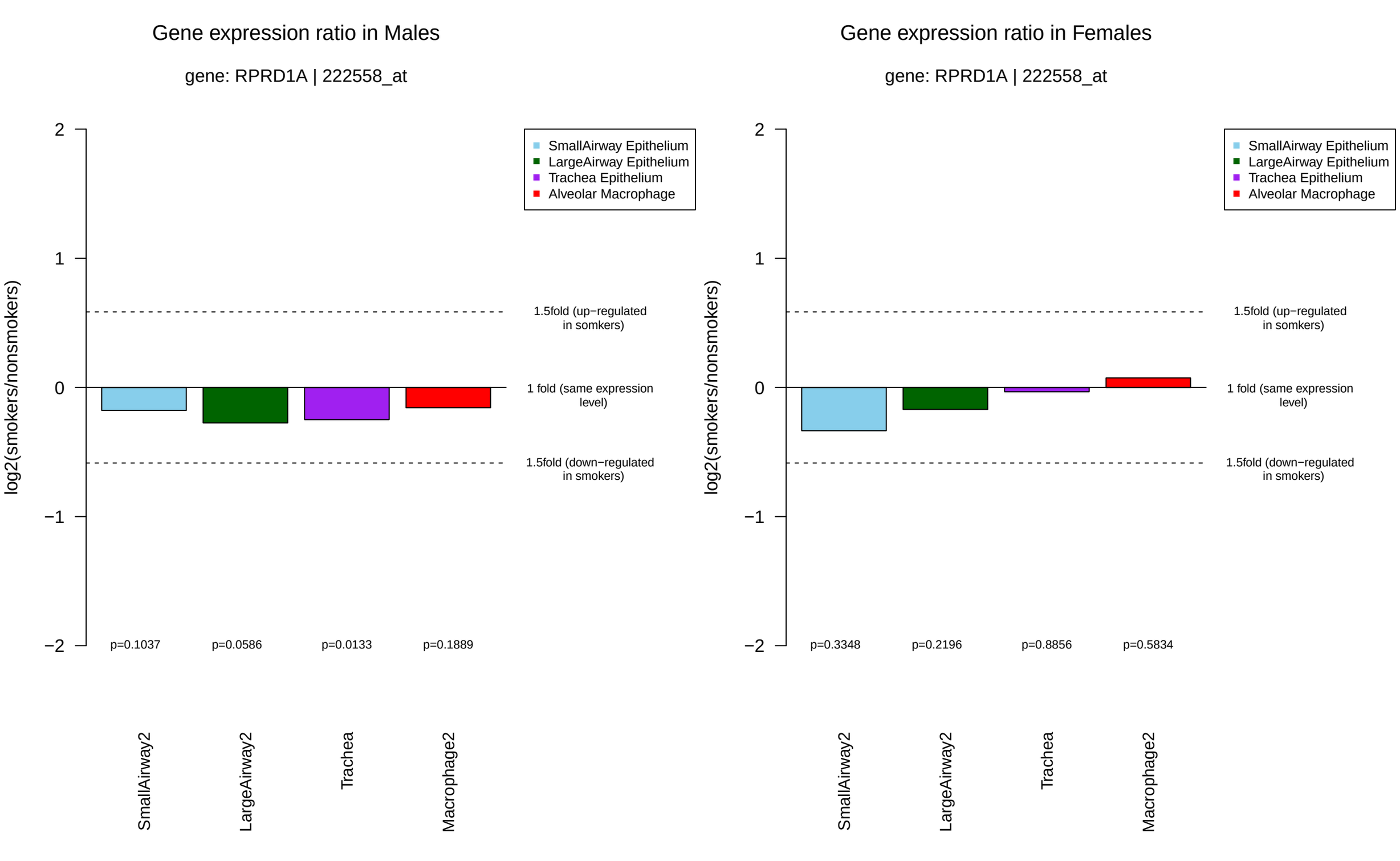
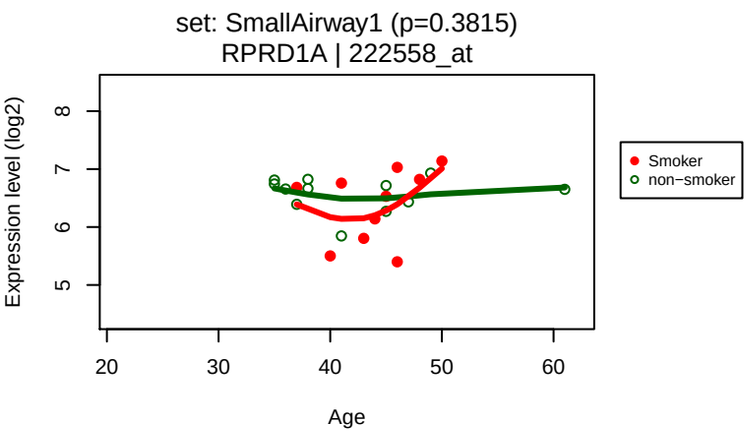
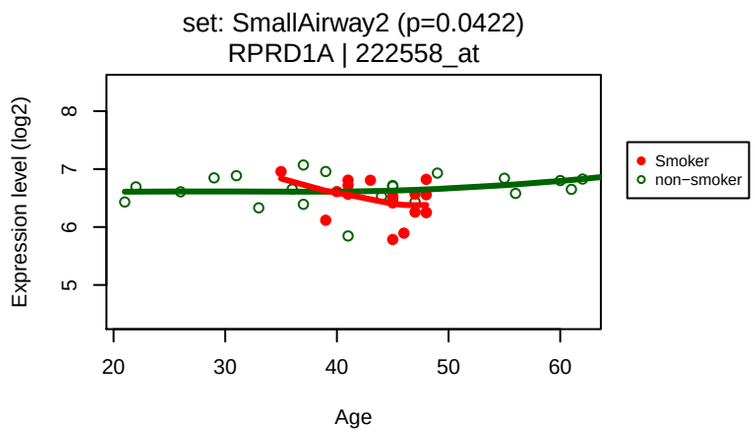
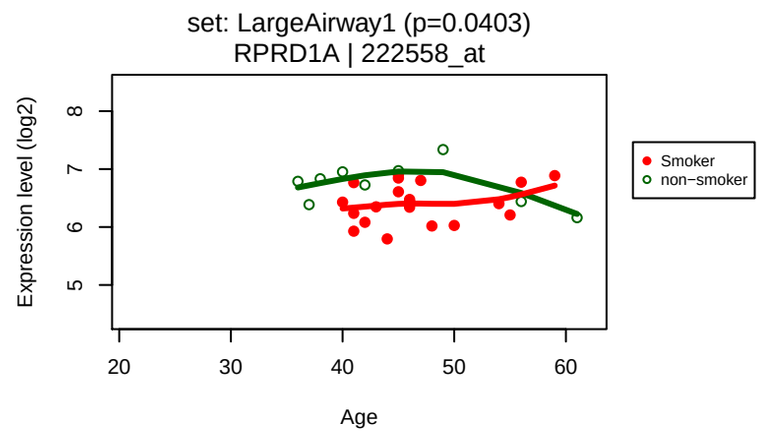

Gene expression of smokers and nonsmokers
gene: RPRD1A | 222558_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3815
p=0.0422
p=0.0015
p=0.0403
p=0.0202
p=0.0271
p=0.7438
p=0.3722
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

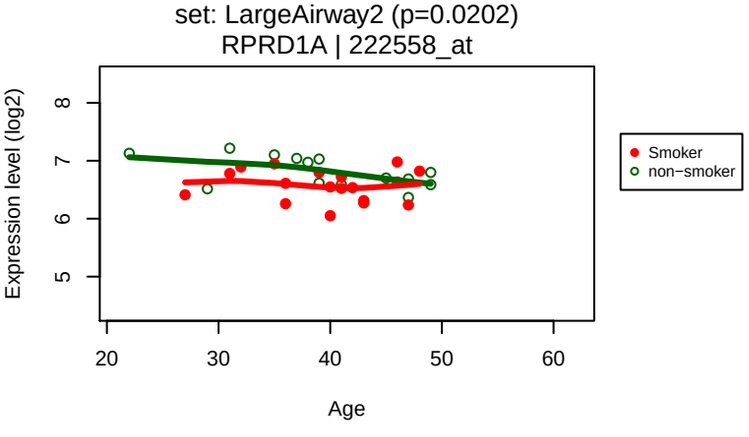

Gene expression ratio (smokers/nonsmokers)
gene: RPRD1A | 222558_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3815
p=0.0422
p=0.0015
p=0.0403
p=0.0202
p=0.0271
p=0.7438
p=0.3722
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

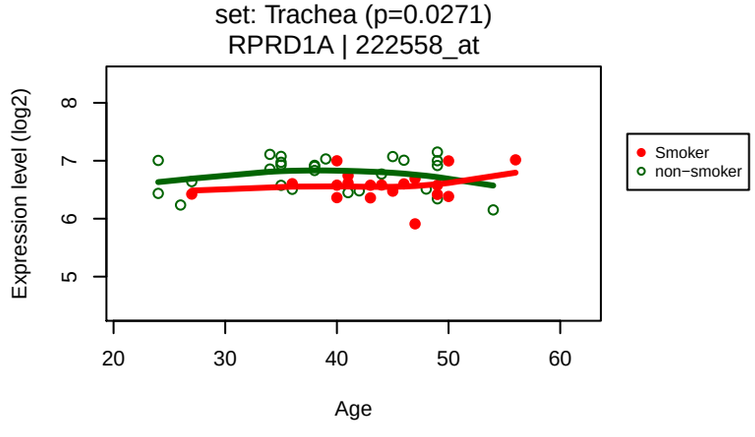

Gene expression ratio in Males
gene: RPRD1A | 222558_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1037
p=0.0586
p=0.0133
p=0.1889
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPRD1A | 222558_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3348
p=0.2196
p=0.8856
p=0.5834
SmallAirway2
LargeAirway2
Trachea
Macrophage2

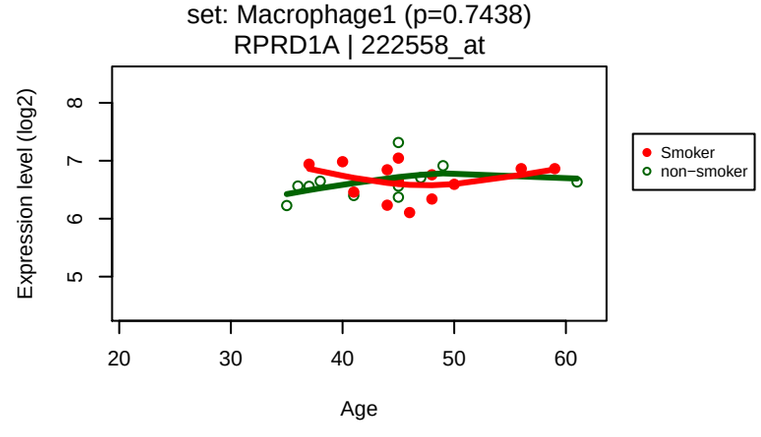

set: SmallAirway1 (p=0.3815)
RPRD1A | 222558_at
Expression level (log2)
Age
Smoker
non−smoker

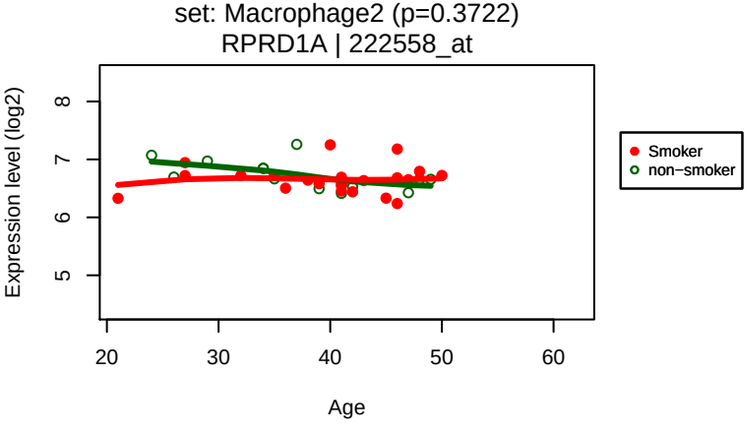

set: SmallAirway2 (p=0.0422)
RPRD1A | 222558_at
Expression level (log2)
Age
Smoker
non−smoker

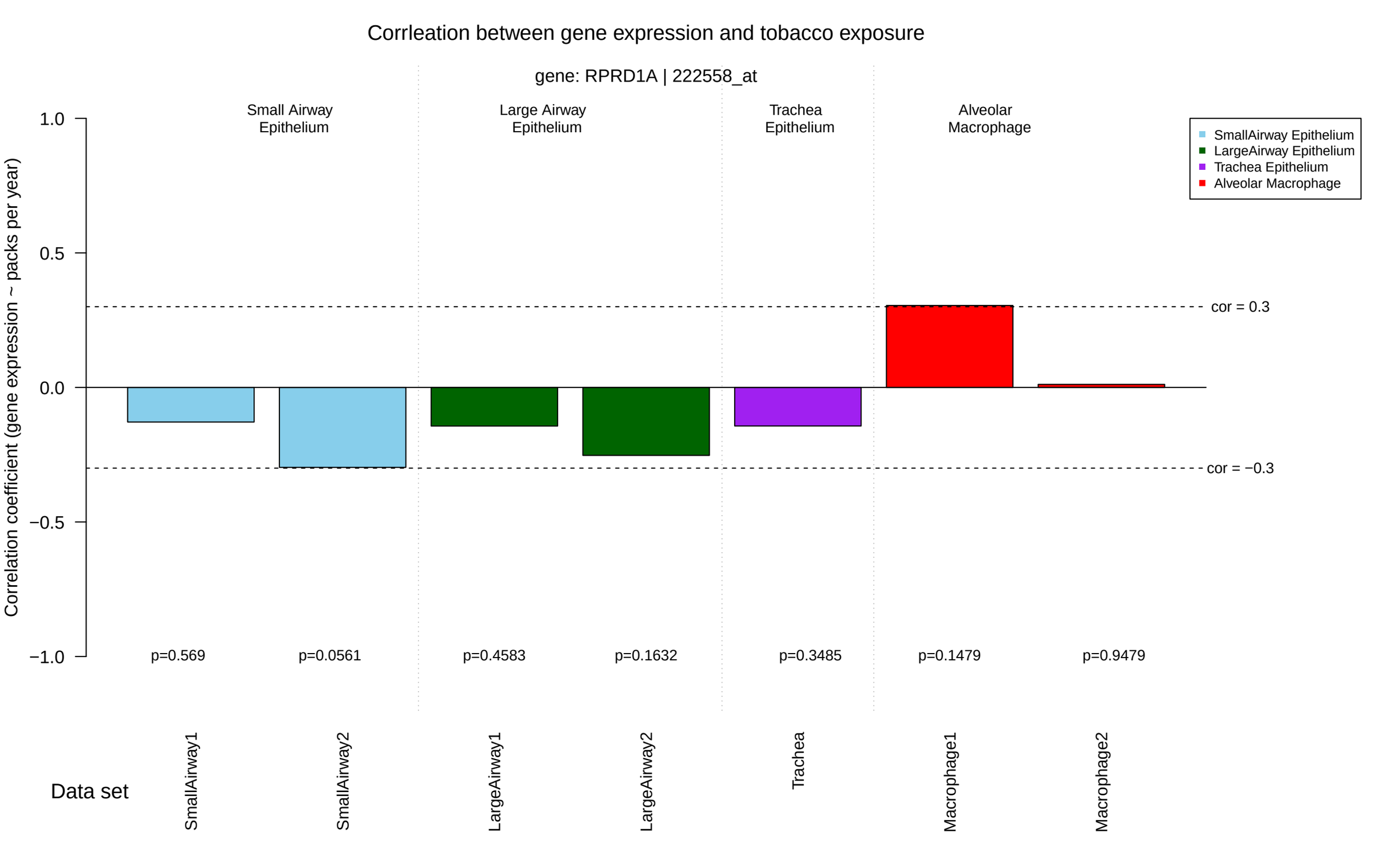

set: LargeAirway1 (p=0.0403)
RPRD1A | 222558_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0202)
RPRD1A | 222558_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0271)
RPRD1A | 222558_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7438)
RPRD1A | 222558_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3722)
RPRD1A | 222558_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPRD1A | 222558_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.569
p=0.0561
p=0.4583
p=0.1632
p=0.3485
p=0.1479
p=0.9479
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2